

Low-voltage control of the magnetoelectric antiferromagnet in Pt/Cr₂O₃/Ru/Pt thin film utilizing the interface lattice misfit

Yoshito Matsumoto¹, Hiroki Sameshima¹, Nozomi Murayama¹, Kentaro Toyoki^{1,2,3},
Ryoichi Nakatani^{1,2,3} and Yu Shiratsuchi^{1,2,3}

¹Graduate School of Engineering, The University of Osaka, Suita 565-0871, Japan, yoshito.matsumoto@mat.eng.osaka-u.ac.jp

²Center for Spintronics Research Network, The University of Osaka, Toyonaka 560-8531, Japan

³Open and Transdisciplinary Research Initiative, The University of Osaka, Suita 565-0871, Japan, shiratsuchi@mat.eng.osaka-u.ac.jp

Antiferromagnets (AFMs) are promising candidates due to robustness against magnetic field perturbation and potential THz manipulation. Magnetoelectric (ME) AFM Cr₂O₃ enables voltage-based switching of AFM spin state via ME effect, holding great promise low-energy consumption. Here, we demonstrate the bipolar AFM spin switching below 200 mV (corresponding to an electric field of 17 mV/nm) using Pt/Cr₂O₃/Ru/Pt thin film. This low-voltage control of AFM spin was achieved by an interface design utilizing the deliberate lattice misfit between Cr₂O₃ and Ru buffer layer. The interfacial lattice mismatch strategy provides a practical pathway to much more ultra-low-power manipulation of AFM spintronic devices.

Index Terms: Antiferromagnetic materials, Low voltage, Magnetoelectric effects, Spintronics

I. INTRODUCTION

Antiferromagnets (AFMs) have attracted significant attention to overcoming the fundamental limitations of conventional ferromagnetic materials. The absence of macroscopic stray fields and the robustness against magnetic field perturbations are suitable for the high-density integration. Moreover, the ultrafast spin dynamics of AFMs are inherent due to the magnetic resonance frequency in THz range, offering a highly attractive pathway to ultrafast data processing. Recent breakthroughs in the electrical manipulation and detection of the AFM order, due to spin transfer torque (STT) [1]-[2] or spin-orbit torque (SOT) [3], have dramatically accelerated these AFM materials from merely fundamental interest to practical device engineering. However, STT and SOT switching still require the huge current density, which lead to high-power consumption due to the undesirable Joule heating.

To achieve the low-power spin manipulation, voltage-driven control has emerged as a compelling alternative to current-induced mechanisms. Magnetoelectric (ME) effect offers a stable mechanism for the deterministic AFM spin switching. In linear ME materials, the applied electric field $\mathbf{E}$ induces a net magnetization ($\mathbf{M} = \alpha \mathbf{E}$, where α is the ME tensor). This cross-coupling enables the direct manipulation of AFM domains despite the absence of a spontaneous magnetization at zero $\mathbf{E}$.

In this work, we focus on Cr₂O₃, a prototypical ME AFM material that exhibits a linear ME effect at room temperature [4]-[5]. In thin film heterostructures, symmetry-breaking at the Cr₂O₃(0001) surface leads to the emergence of boundary magnetization (BM) [6], a sizable spin-polarized surface state. Based on coupling between the BM and the bulk Néel vector, the AFM spin states can be read out via the spin-dependent transport measurements such as the anomalous Hall effect (AHE) at the nonmagnetic heavy metal (HM)/ME AFM interface [7]. We recently reported the voltage-induced switching of the AFM spin state, and also revealed that

suppressing defect-induced emergent magnetization would be beneficial for low-voltage operation. Specifically, structural imperfections like misfit dislocations unbalance the atomic population of spin sublattice, as proposed in ref. 8

To address this challenge, we employed interfacial engineering to control dislocation formation. According to theoretical models, a large lattice mismatch minimizes the critical thickness. Consequently, we promote rapid lattice relaxation at the initial stage of Cr₂O₃ growth by adopting a Ru(0001) buffer layer, which has -5.4 % interfacial lattice misfit with Cr₂O₃(0001). This deliberate introduction of large lattice mismatch, we achieved a low-voltage AFM spin switching in Pt/Cr₂O₃/Ru/Pt epitaxial system.

II. EXPERIMENTAL METHODS

We fabricated Pt(2 nm)/Cr₂O₃(10 nm)/Ru(15 nm)/Pt(5 nm) multilayered film on an α -Al₂O₃ (0001) substrate by DC magnetron sputtering. First, a 5-nm Pt layer was formed at 873 K, followed by a fabrication of a Ru layer at room temperature. Then, the film was annealed at 773 K for 30 minutes to smoothen the surface. The Cr₂O₃ layer was fabricated at 773 K using reactive sputtering with a pure Cr target in an atmosphere of a mixture of Ar and O₂. Finally, a 2-nm layer of Pt was deposited at room temperature.

The crystalline quality and epitaxial relationship were characterized by using reflection high-energy electron diffraction (RHEED) and X-ray diffraction (XRD). The RHEED images for each layer were obtained at the acceleration voltage of 25 kV and the emission current of 50 μ A. The XRD measurements were done using Cu $K\alpha$ irradiation at the acceleration voltage and the emission current of 45 kV and 200 mA, respectively.

In the AHE measurements, the film was patterned into a micro-Hall bar device (width: 5 μ m, length: 20 μ m) by photolithography and Ar ion milling. SiO_x layer was deposited around the device to ensure electrical insulation for gate voltage

application. The gate voltage was applied between the top 2-nm Pt and the bottom Ru/Pt layers. AHE was measured using the fabricated device with a lock-in amplifier. A 100 μA and 1 kHz electric current was applied to the upper electrode, and the Hall voltage (ΔV_{xy}) was measured as a function gate voltage (V_g) upon applying static magnetic field. V_g was swept to the Cr_2O_3 layer in the direction perpendicular to the film within a range of ± 300 mV. The measurement temperature was 270 K, slightly below the Néel temperature (T_N) of the fabricated film, 274 K which was determined by the temperature dependence of AHE conductivity for the identical film used for the ME measurements.

III. RESULTS AND DISCUSSION

A. Structural Characterizations

Figure 1 (a)-(d) show the RHEED images for Cr_2O_3 and Ru buffer layer for the different electron azimuth, respectively. The sharp streaked patterns indicate the atomically flat surface and epitaxial growth. In the patterns of Cr_2O_3 , the symmetric diffraction patterns suggest the presence of twin crystal. In the case of Ru, the streaks between the fundamental diffraction are observed, which is attributed to the surface reconstruction due to the oxygen adsorption on the Ru surface [9]. Combined with the XRD measurements, we determined the epitaxial orientation: $\text{Cr}_2\text{O}_3(0001)[11\bar{2}0] \parallel \text{Ru}(0001)[1\bar{1}00]$. A key highlight of our structural investigation is the role of the lattice misfit (-5.4%). While such a large misfit might conventionally be expected to hinder epitaxial growth, our RHEED data indicates that the epitaxial growth persisted despite the large lattice mismatch.

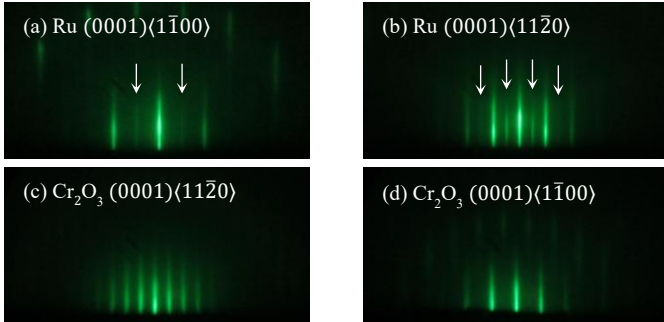


Fig. 1. RHEED images for Ru and Cr_2O_3 layer. The white arrows in (a) and (b) point to the streaks caused by the surface reconstruction.

B. Electric and Magnetic properties

We explored the voltage-driven switching of the Néel vector as the targeting functionality of our Pt/ Cr_2O_3 /Ru/Pt heterostructure. Figure 2 shows the AHE loops as a function of V_g measured at 270 K under a static magnetic field $\mu_0 H$. The clear hysteresis was observed, indicating that the AFM spin state was switched by the voltage sweep. The switching event occurred at V_g of approximately 0.2 V corresponding to an electric field of 17 mV/nm, which confirms the high efficient voltage-induced switching. The fact that the $\mu_0 H$ value is not so high, i.e., ± 1 T is also important advance compared with the previous ME materials. Further, we revealed, using the simultaneously collected I - V curve across the Cr_2O_3 layer, the

leakage current in the switching condition was about 21 nA (1.4×10^2 A/m²) at 0.2 V.

In previously reported Cr_2O_3 thin film [8], [10], the emergent magnetization due to the structural imperfections often caused a difference of the switching voltage depending on the switching direction. The results shown in Fig. 2 indicate that the V_g values for both reversals are approximately the same, suggesting the absence of offset voltage. In our system, we attribute the suppression of this offset to the deliberate utilization of the Ru buffer's large misfit, as discussed above.

In summary, despite using the Ru buffer having the large lattice mismatch with Cr_2O_3 , we have obtained the epitaxially grown ME AFM $\text{Cr}_2\text{O}_3(0001)$ thin film. Using the fabricated film, we demonstrated the the energy-efficient AFM spin switching. The switching voltage could be below 200 mV, and the quite small offset voltage, about 3.5 mV/nm which is one-order lower than other systems [10]. The HM/ME-AFM system as a viable platform for the next generation of low-power, non-volatile memory and logic devices.

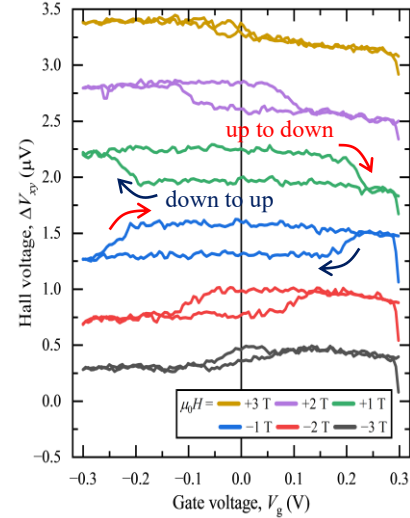


Fig. 2. AHE loops measured at 270 K under static $\mu_0 H$.

REFERENCES

- [1] J. C. Slonczewski, "Current-driven excitation of magnetic multilayers," *J. Magn. Magn. Mater.* vol. 159, pp. L1-L7, Jun. 1996.
- [2] D. C. Ralph et al., "Spin transfer torques," *J. Magn. Magn. Mater.* vol. 3210, pp. 1190-1216, Apr. 2008.
- [3] P. Wadley et al., "Electrical switching of an antiferromagnet," *Science*, vol. 351, pp. 587-591, Jan. 2016.
- [4] D. N. Astrov, "Magnetoelectric effect in chromium oxide," *J. Exp. Theor. Phys.* vol. 13, pp. 729-733, Oct. 1961.
- [5] G. T. Rado, "Mechanism of the magnetoelectric effect in an antiferromagnet," *Phys. Rev. Lett.* Vol. 6, pp. 609-610, Jun. 1961.
- [6] K. D. Belashchenko, "Equilibrium magnetization at the boundary of a magnetoelectric antiferromagnet," *Phys. Rev. Lett.* vol. 105, 147204, Oct. 2010.
- [7] T. Kosub et al., "All-electric access to the magnetic-field-invariant magnetization of antiferromagnets," *Phys. Rev. Lett.* vol. 115, 097201, Aug. 2015.
- [8] T. Kosub et al., "Purely antiferromagnetic magnetoelectric random access memory," *Nature Comms.* vol. 8, Jan. 2017.
- [9] H. Pfnür et al., "Oxygen induced reconstruction of a close-packed surface: a LEED IV study on $\text{Ru}(001)\text{-p}(2\times 1)\text{O}$," *Surf. Sci.* vol. 220, pp. 43-58, May 1989.
- [10] H. Sameshima et al., "Highly efficient deterministic voltage control of the perpendicular Néel vector in a magnetoelectric antiferromagnet," *Phys. Rev. Applied* vol. 25, 014017, Jan. 2026.